

NPTEL Course on "Computer System Optimization for ML"

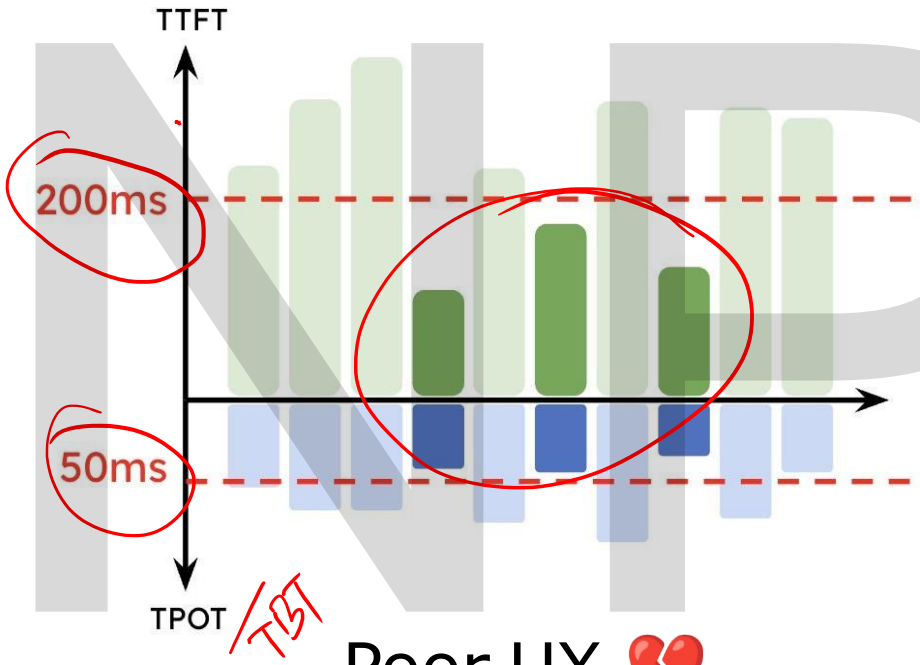
Lecture #34

Agenda

- Distributed Service (contd.)
- Fault Tolerance in Machine Learning

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Colocation → Overprovision Resource to meet SLO

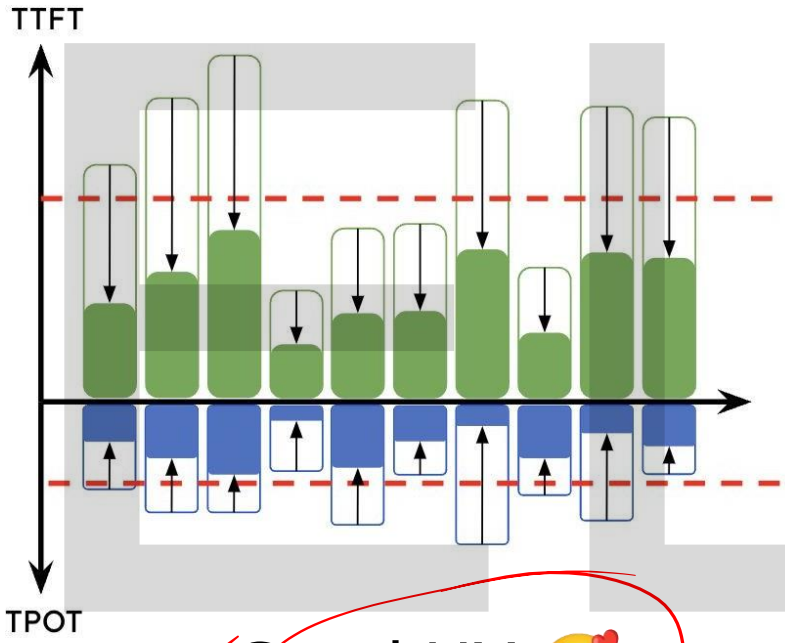


Poor UX 💔

lower cost 💰

GPU x1 → GPU GPU GPU GPU x4

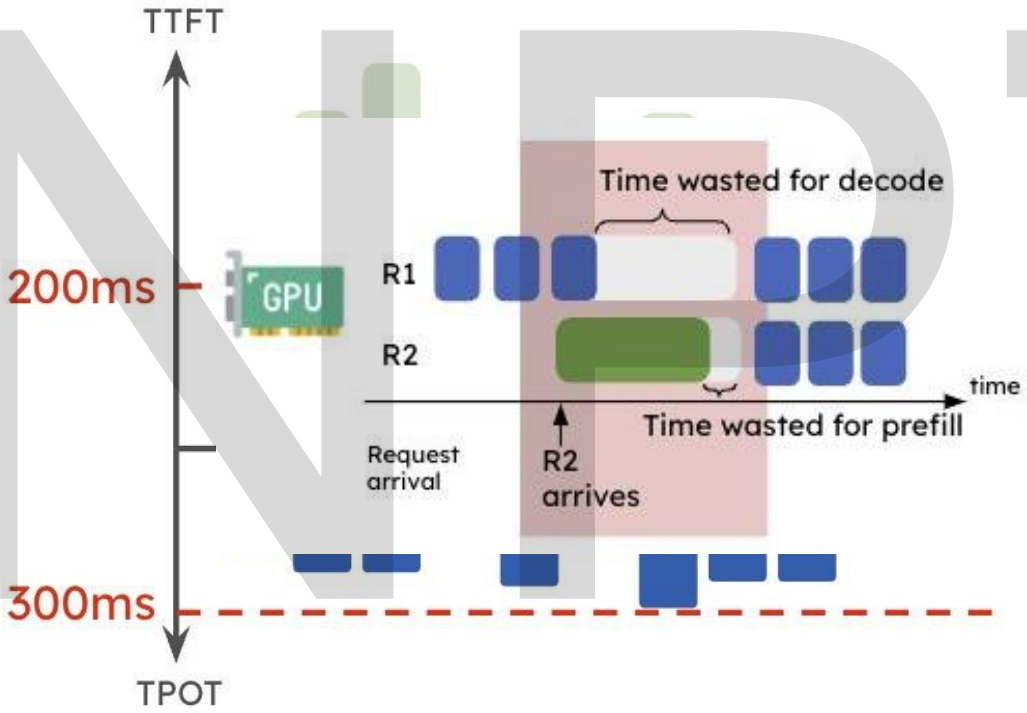
add more GPU



Good UX 🥰

Higher cost 💰💰💰💰

Colocation → Coupled Parallelism

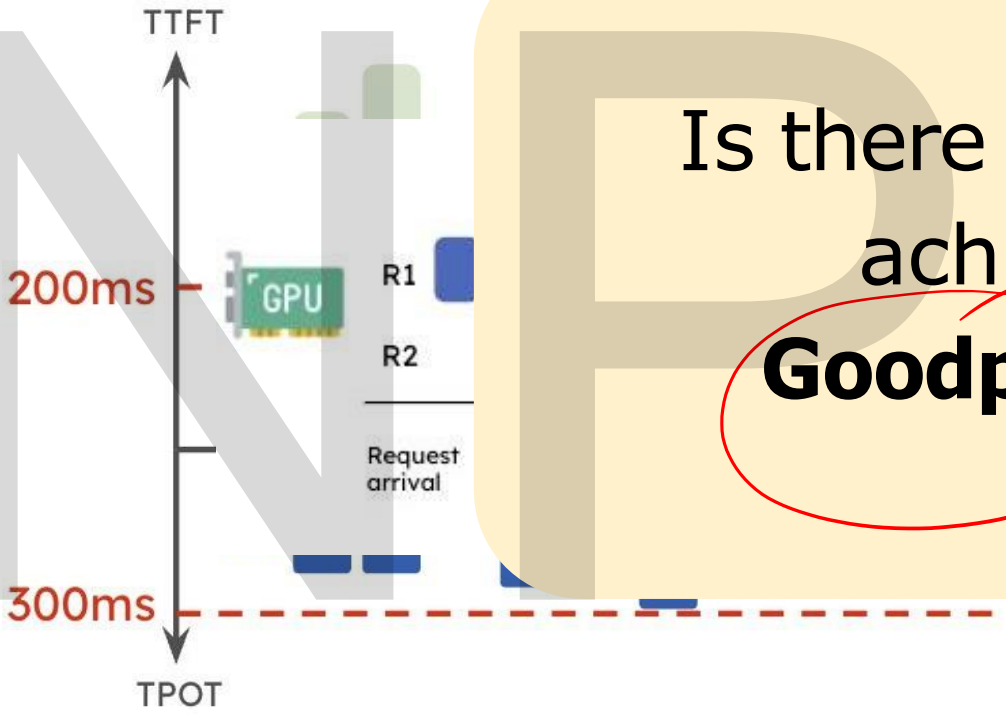


TTFT tight, TPOT loose

	PP	TP
• Prefill	😐	❤️
• Decode	❤️	😐

Prefill and Decode have different preferences

Colocation → Coupled Parallelism

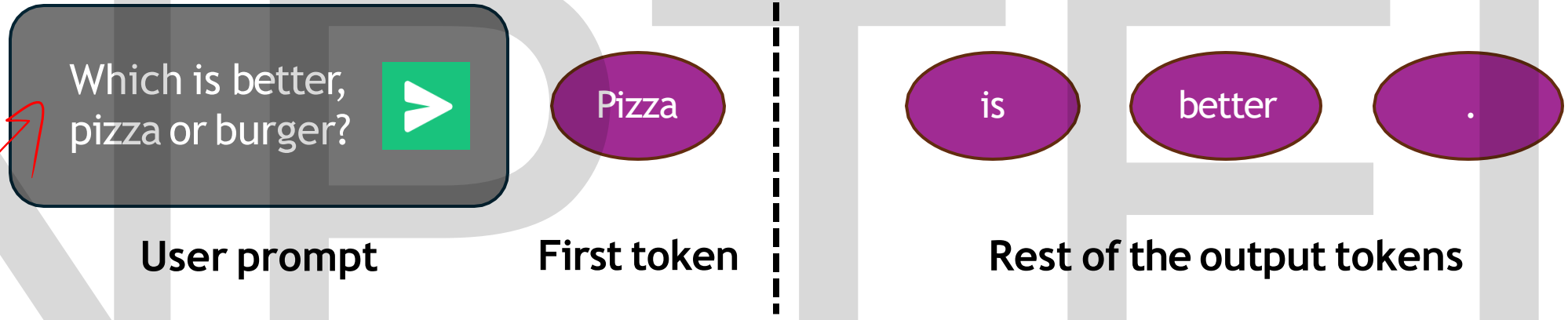


Is there a better way to achieve better **Goodput per GPU**?

TTFT tight, TPOT loose

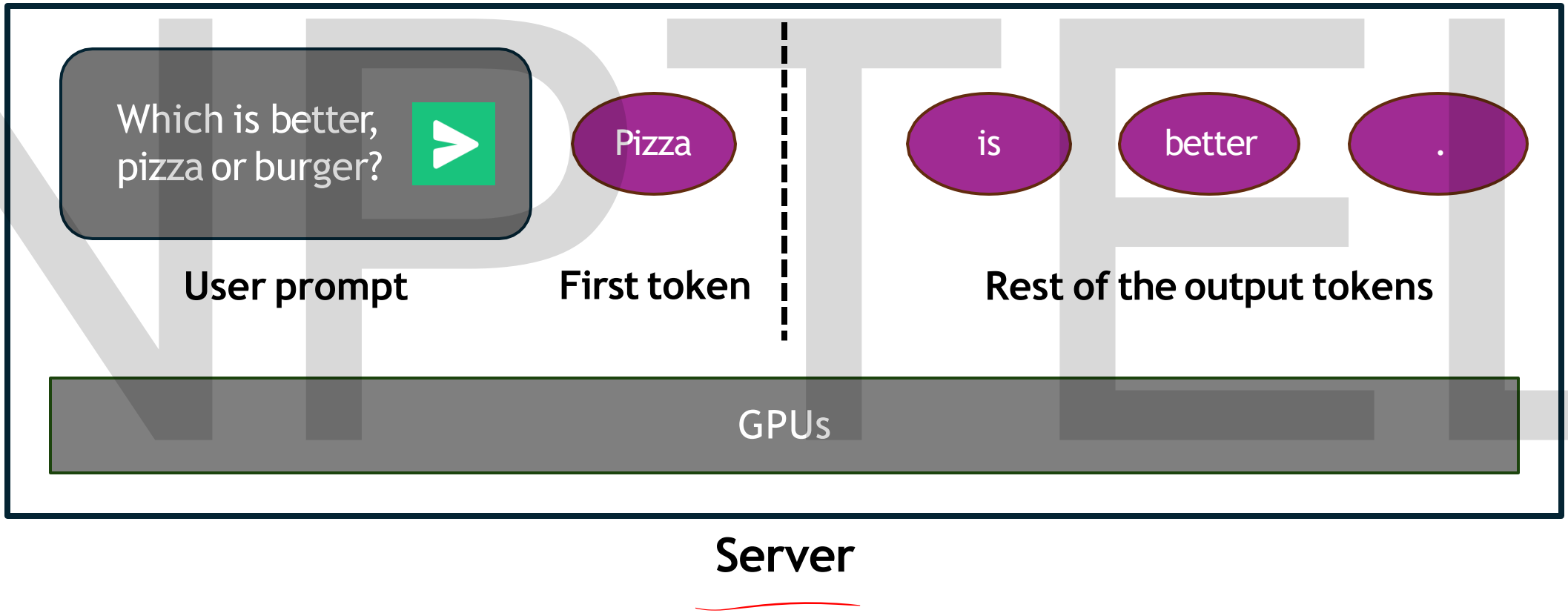
Prefill and Decode have different preferences

Prompt computation vs. token generation



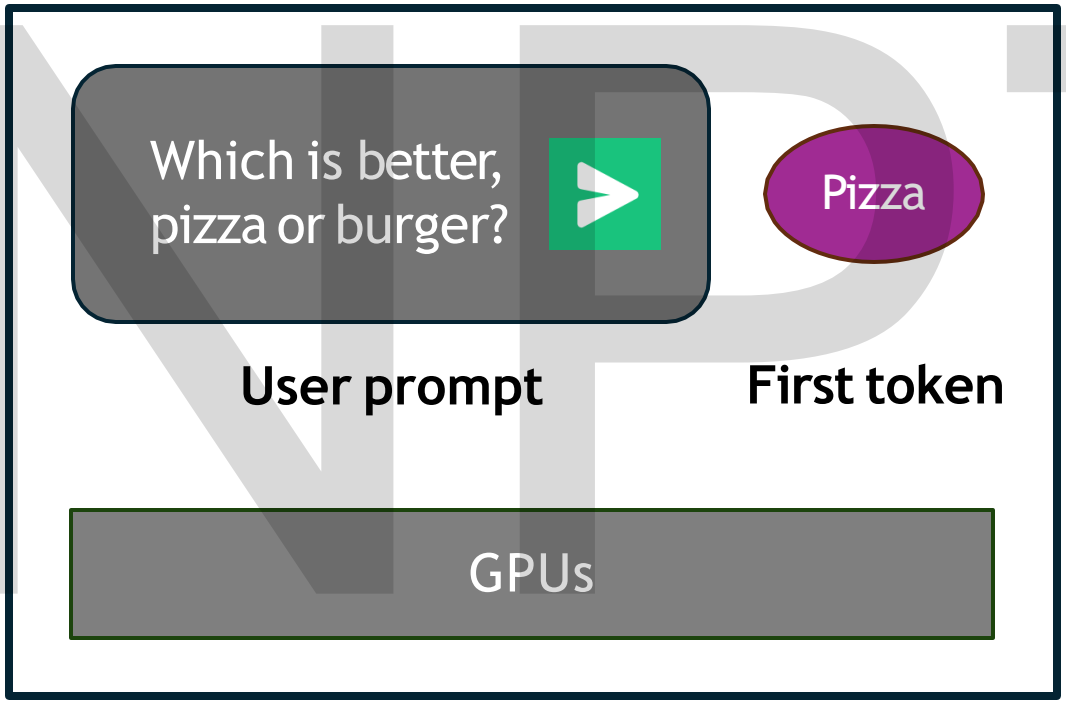
Prompt phase	Token phase
Compute and power intensive	Memory intensive
Limited batching benefits	Batching improves throughput

Inefficient to run both on the same hardware



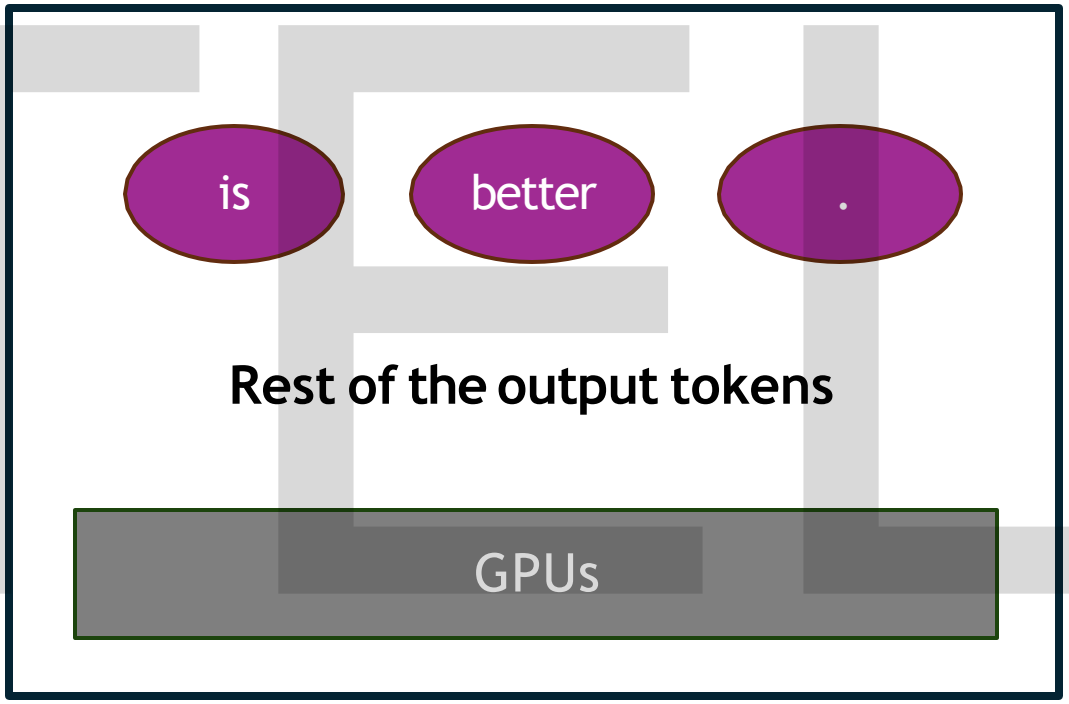
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Splitwise/Distserve splits phases onto different servers



Prompt server

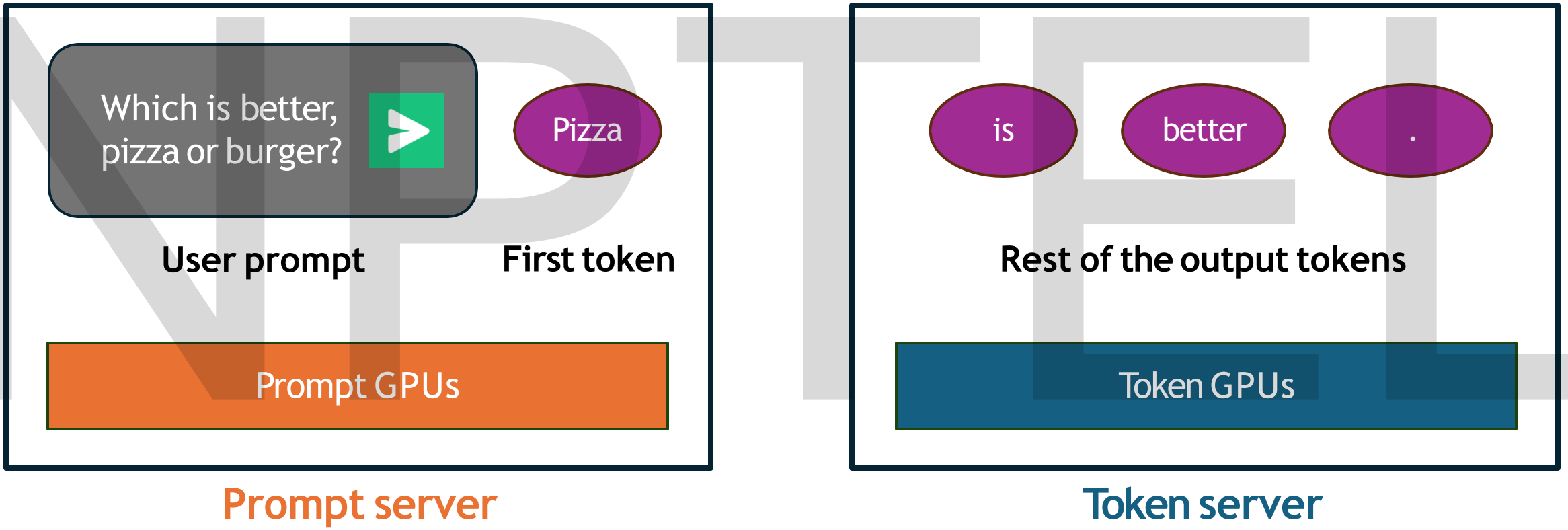
Small batches, maximum power



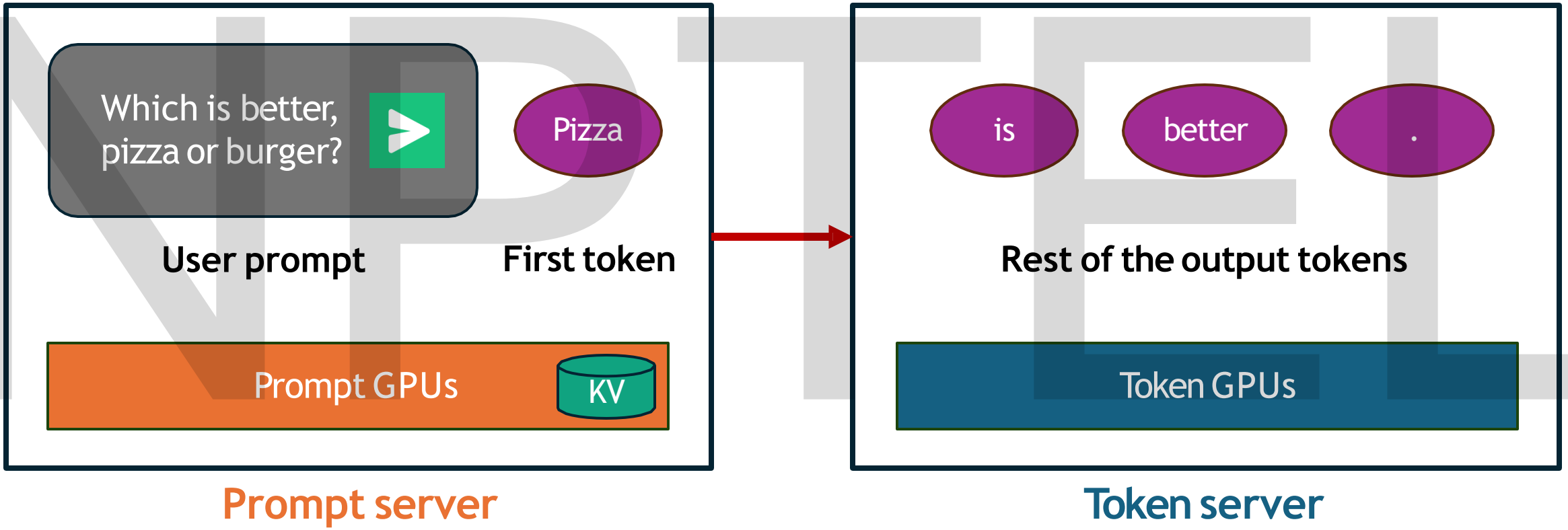
Token server

Large batches, power capped

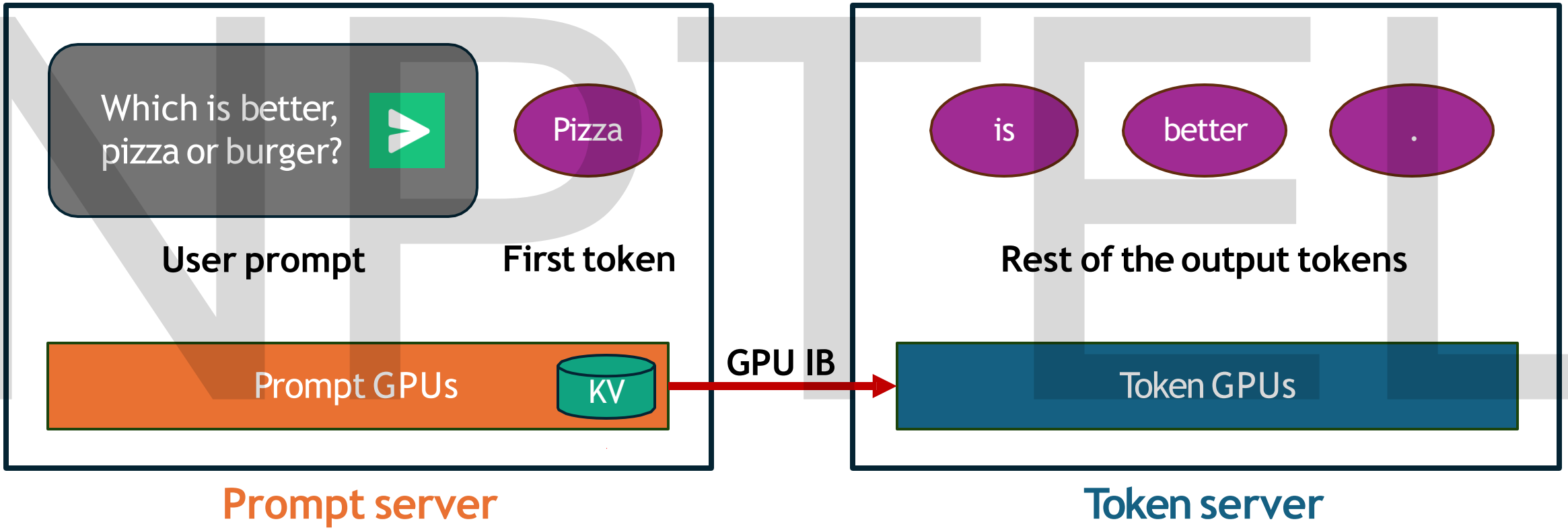
Each phase could use preferred hardware



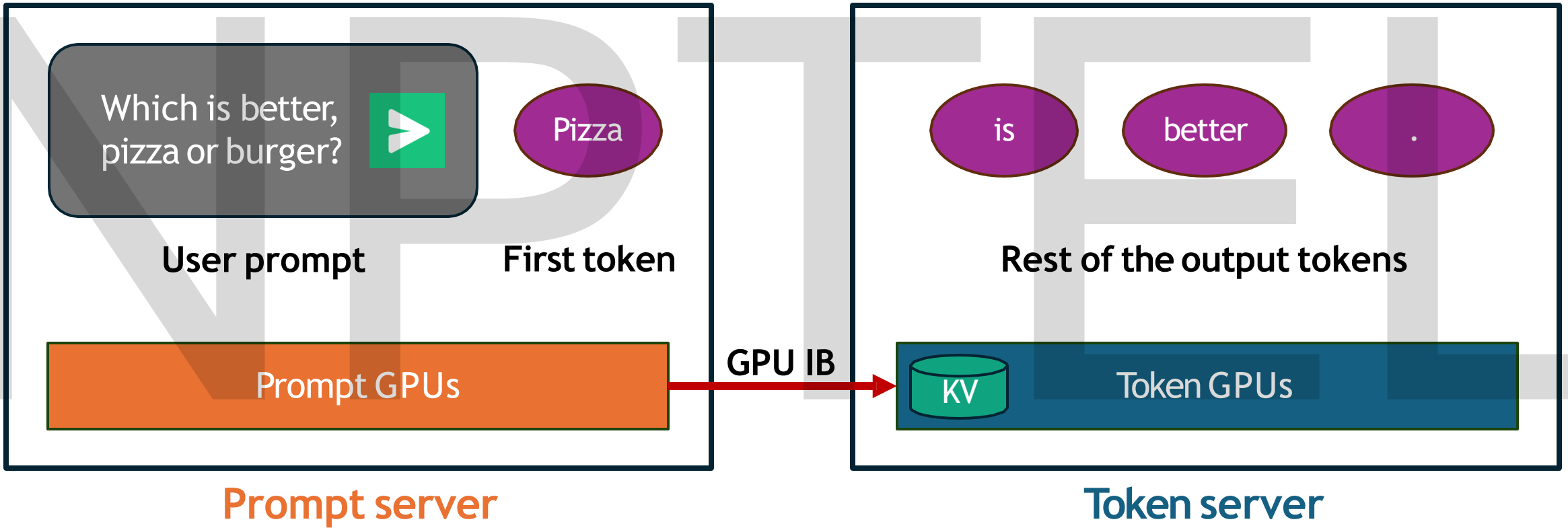
Splitting inference requires fast state transfers



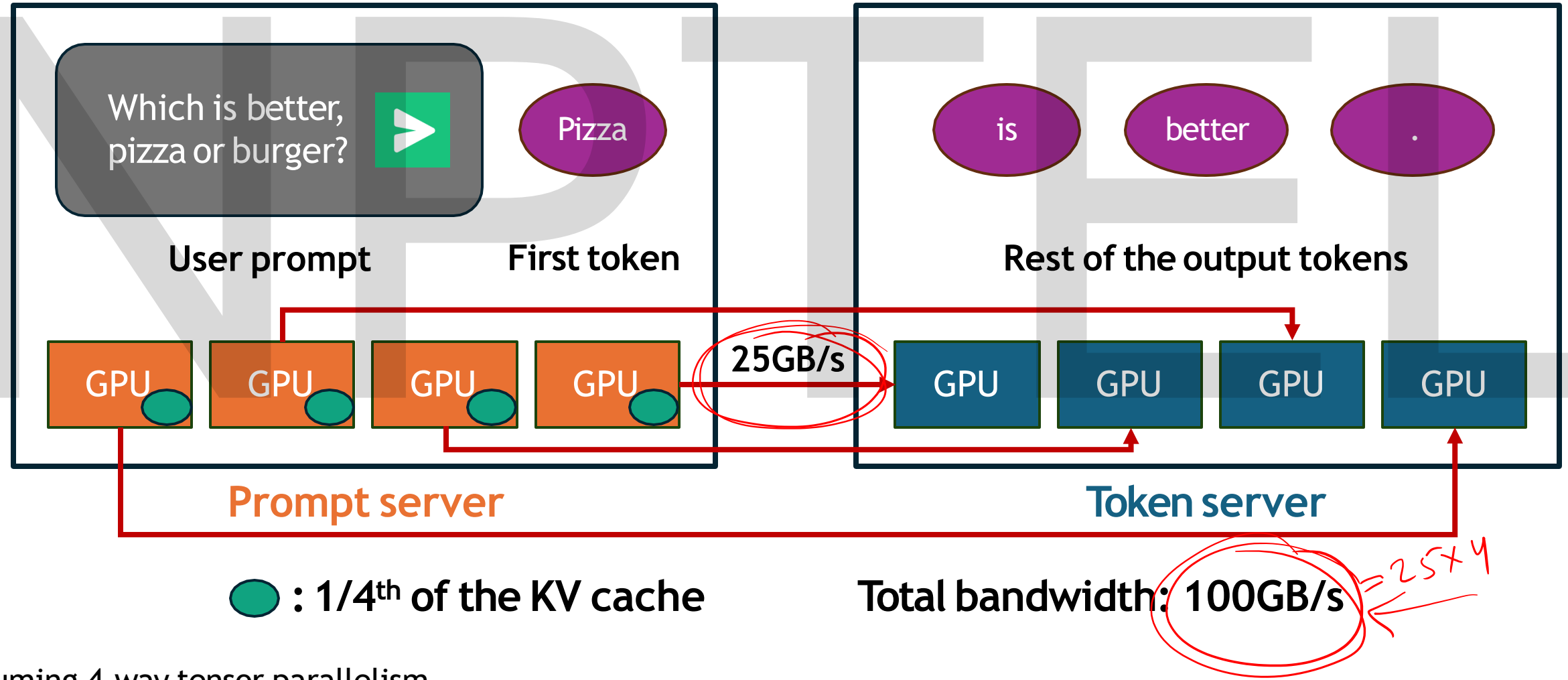
Splitwise uses GPU Infiniband to ship state



Splitwise uses GPU Infiniband to ship state

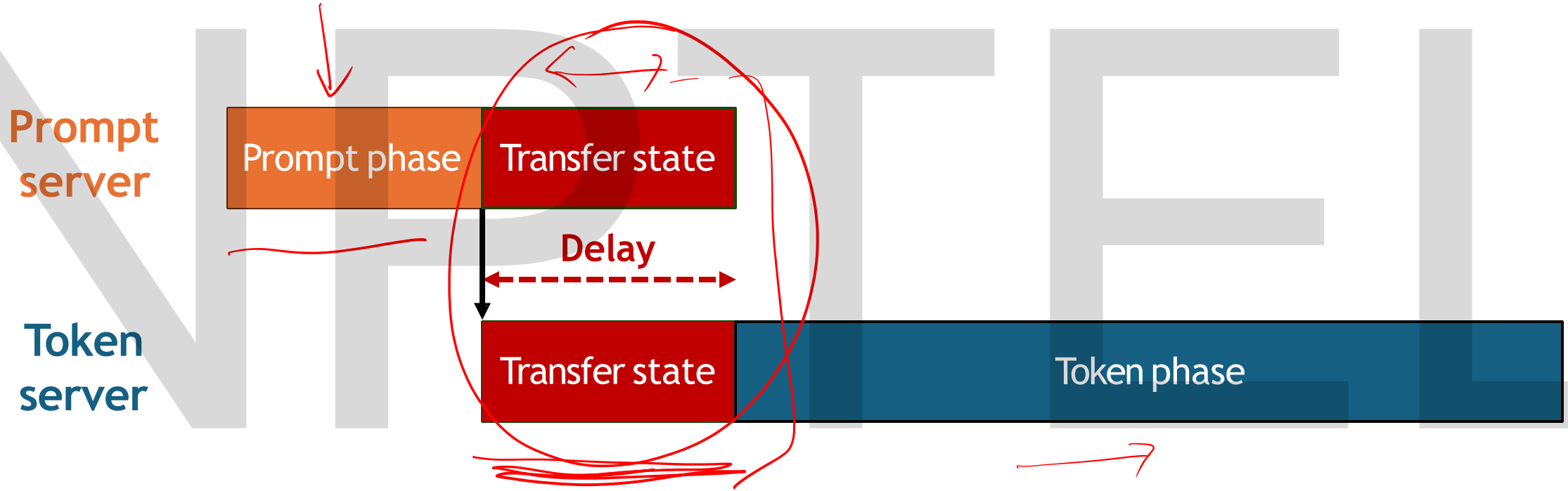


Parallelize transfers for high bandwidth



Assuming 4-way tensor parallelism

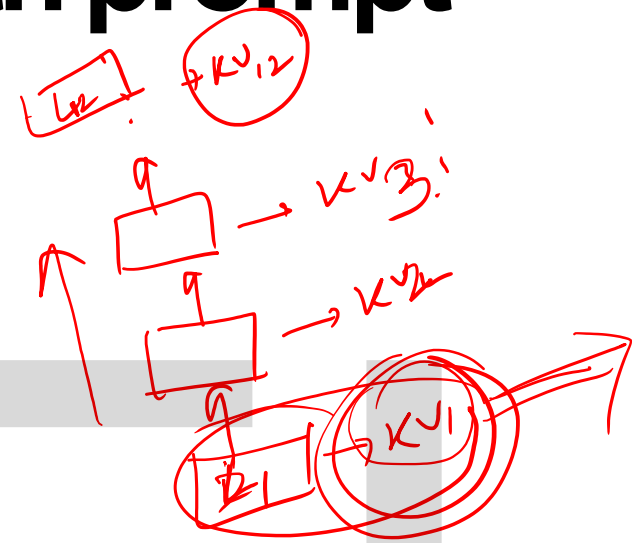
Transfer overheads may still be large

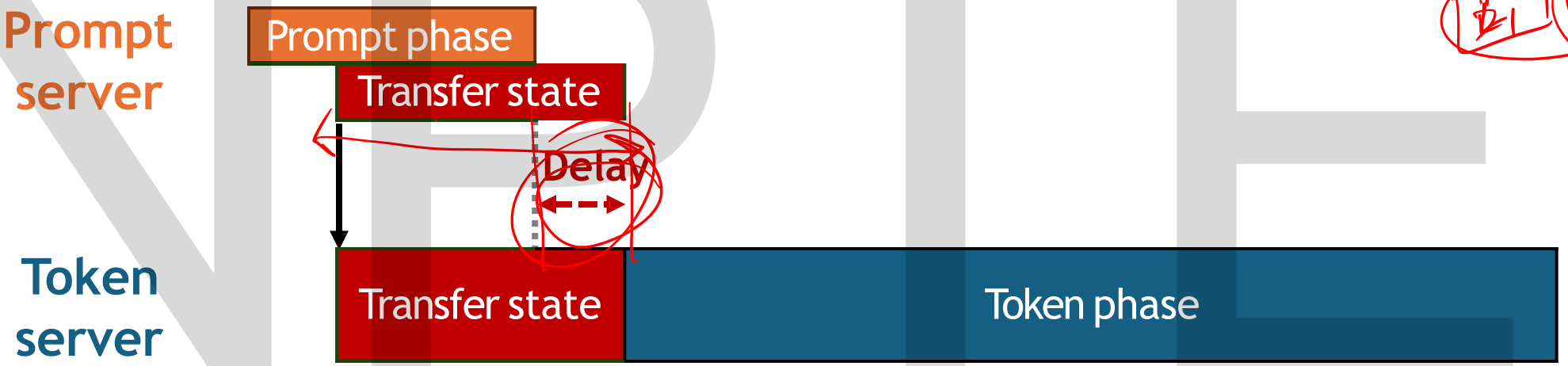


KV cache sizes can be hundreds of GBs!

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Splitwise overlaps transfer with prompt phase





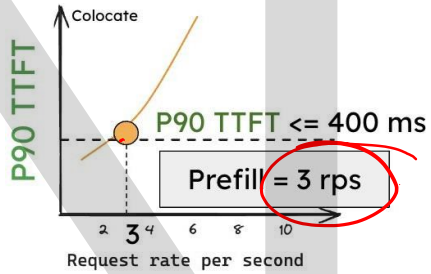
Start shipping the KV-cache after the first prompt layer

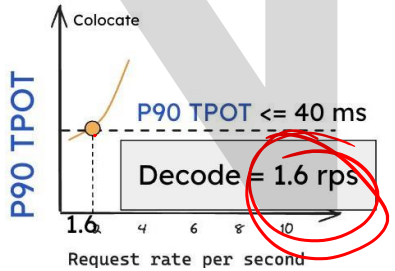
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Disaggregation achieves better goodput

Colocate

1 GPU for both Prefill and Decode







Max System goodput

= Min(Prefill, Decode)

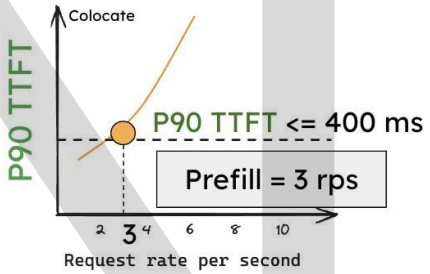
= 1.6 rps / GPU 😐

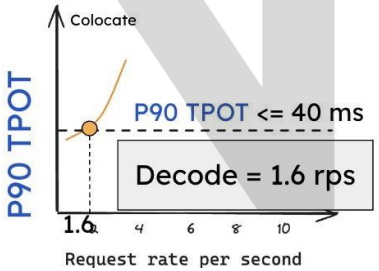
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Disaggregation achieves better goodput

Colocate

1 GPU for both Prefill and Decode







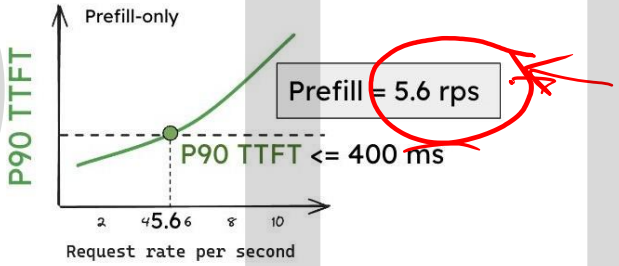
Max System goodput

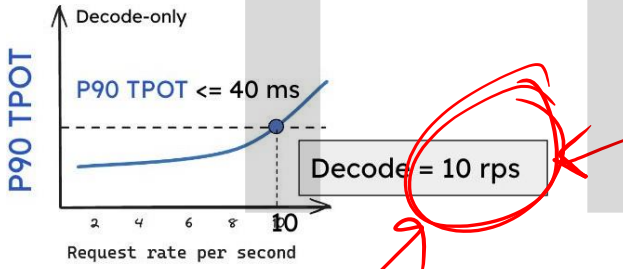
= Min(Prefill, Decode)

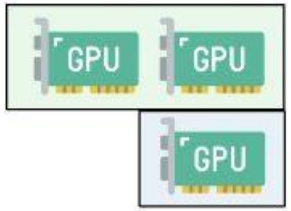
= 1.6 rps / GPU 😐

Disaggregate (2P1D)

2 GPU for Prefill + 1 GPU for Decode







Disaggregate (2P1D) goodput

= Min (5.6 x 2, 10) rps / 3 GPU

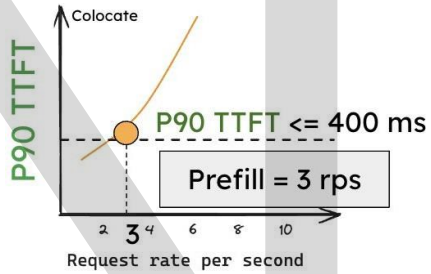
= 3.3 rps / GPU 😎

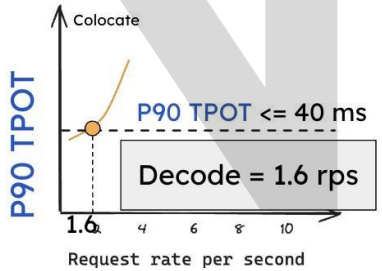
@10 RPS

Disaggregation achieves better goodput

Colocate

1 GPU for both Prefill and Decode







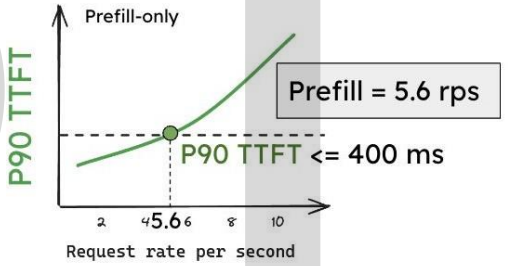
Max System goodput

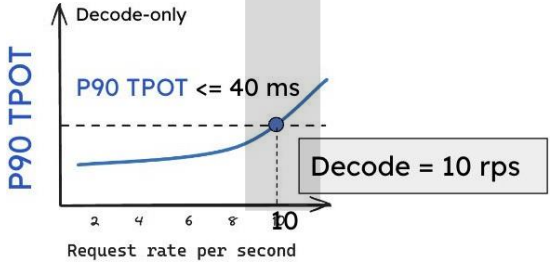
= Min(Prefill, Decode)

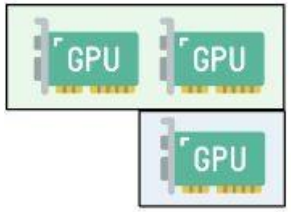
= 1.6 rps / GPU 😐

Disaggregate (2P1D)

2 GPU for Prefill + 1 GPU for Decode







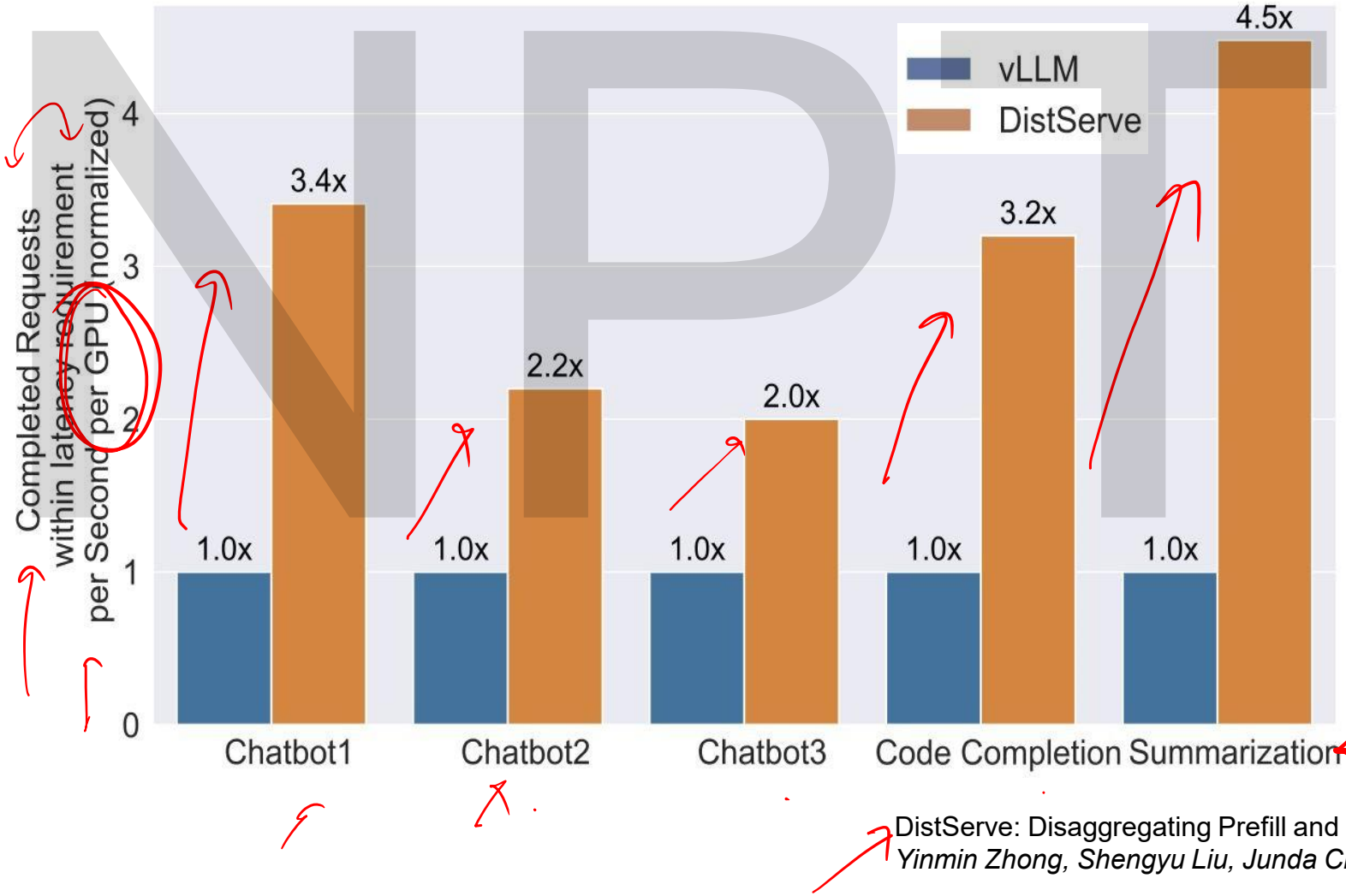
Disaggregate (2P1D) goodput

= Min (5.6 x **2**, 10) rps / 3 GPU

= 3.3 rps / GPU 😎

Simple Disaggregation achieves **2x** goodput (per GPU)

Evaluation



Achieves 2.0x - 4.48x compared to vanilla vLLM

- Chatbot: 2.0 - 3.4x
- Code Completion: 3.2x
- Summarization: 4.5x

DistServe: Disaggregating Prefill and Decoding for Goodput-optimized Large Language Model Serving
Yinmin Zhong, Shengyu Liu, Junda Chen, Jianbo Hu, Yibo Zhu, Xuanzhe Liu, Xin Jin, Hao Zhang



THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)